

Methodology and Limitations

Naqsha Collective: Garden East Waste Site Analysis

Map 1: Composite Vulnerability Index

Methodology

This map extends the waste site analysis into a preliminary composite vulnerability index, combining two indicators: population density and waste-site risk. Population density data was sourced from WorldPop (2020, 1km resolution, UN-adjusted). Waste-site risk was represented using the tier-weighted density surface described above.

Both layers were normalised to a common 0–1 scale using min-max normalisation, so that each indicator contributes on an equivalent range regardless of its original units. Because the two datasets differed substantially in spatial resolution, they were aligned to a matching extent and cell size before being combined. The normalised layers were then combined using equal weighting (each contributing 50 percent to the final score), producing a single composite raster in which higher values indicate a greater overlap of population density and waste-site risk.

Limitations

Equal weighting was used as a starting assumption rather than one derived from statistical or expert-informed justification; future iterations could explore differential weighting once additional indicators are available. The population layer's 1km resolution is considerably coarser than the site-level waste data, meaning the composite can indicate general locational context but cannot yet distinguish fine-grained variation within a single population grid cell. The index currently includes only two indicators; a fuller vulnerability index would benefit from the addition of further layers, such as flood risk, income proxies, and drainage proximity, which were not available in a usable form within the current project timeline. As with Map 1, the study area remains limited to the Garden East pilot site.

Next Steps

Planned extensions include incorporating flood or elevation-based hazard data, testing differential indicator weighting, and expanding the underlying field dataset to additional neighbourhoods to allow comparative analysis across sites.

Map 2: Waste Site Toxicity and Density Map

Methodology

This map documents the spatial distribution and relative hazard of informal waste sites in Garden East, Karachi, using field data collected by Naqsha Collective. Each site was visited in person and classified using two independent systems: a Toxicity Tier (based on visible material indicators, following the Visual Identification Protocol) and a Burning Status. Site locations were recorded using GPS coordinates.

To identify areas of concentrated risk, a kernel density estimation (heatmap) was generated from the eight recorded sites in QGIS. Each site was assigned a numeric weight based on its toxicity tier (Tier 1 = 1, Tier 2 = 2, Tier 3 = 3), so that higher-hazard sites contribute more strongly to the resulting density surface than lower-hazard ones. The output was symbolized on a graduated colour scale, with individual site points overlaid and coloured by tier for reference.

Limitations

The dataset consists of eight documented sites within a single pilot area and does not yet represent a city-wide or statistically representative sample. Toxicity tier assignments rely on field-based visual classification, which, while guided by a consistent protocol, retains an element of observer judgement. The kernel density radius and pixel size were chosen through iterative visual inspection rather than a formally optimised bandwidth. Expanding the number of sites and pilot areas is a planned next step.